A Real-Time Forecasting-Ensemble Index of Economic Uncertainty and the Volatility of GDP Revisions

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Abstract

Statistical agencies face a fundamental trade-off between timeliness and accuracy, particularly during economic crises when real-time data becomes structurally disordered. We ask whether the dispersion and instability of a real-time forecasting ensemble anticipate the magnitude of subsequent revisions in official GDP figures, and whether this signal adds information beyond established uncertainty proxies. We construct a Composite Economic Uncertainty Index (CEUI) from three real-time dimensions of a five-model ensemble (VAR, Random Forest, ARIMA, LSTM, and Dynamic Factor Models)—within-model variability, between-model dispersion, and temporal instability—applied to Spanish GDP growth over 2015Q4–2024Q4. The index, computed strictly from information available in real time, is strongly associated with the magnitude of subsequent revisions ($\rho = 0.729, p < 0.001$). In a horse-race against the VIX and the Economic Policy Uncertainty (EPU) index for Spain, the CEUI carries predictive information about data quality that the external proxies do not: it remains significant at the 1% level while the VIX and EPU become insignificant once it is included. The results suggest that the internal disorder of a forecasting system is a useful real-time leading indicator of the reliability of official statistics.

JEL codes: C53, C82, E01, E32.
Keywords: Economic Uncertainty, Forecast Combinations, Informational Entropy, GDP Revisions, Real-time Data.

1 Introduction

Modern macroeconomics and policy design rely heavily on the prompt availability of National Accounts. However, statistical agencies such as Spain's INE (*Instituto Nacional de Estadística*) face an inherent trade-off between timeliness and precision (Mankiw and Shapiro, 1986). The production of "flash" GDP estimates requires statistical triangulation under conditions of incomplete information, leading to substantial ex-post revisions as newer and more comprehensive data becomes available. While revisions are a natural part of the statistical production cycle, their magnitude and direction are not uniform over time. As Orphanides (2001) noted, utilizing real-time data that is prone to large

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subsequent revisions can lead to significant policy missteps, particularly in real-time macroeconomic calibration and monitoring.

Recently, global disruptions—the 2008 financial crisis, the European sovereign debt crisis, and the COVID-19 pandemic—have highlighted that this timeliness-accuracy trade-off is drastically amplified during periods of economic stress. In stable regimes, revisions to Spanish GDP growth are relatively small. However, the magnitude of these corrections (Mean Absolute Error, MAE) spikes by amplification factors of 1.5x during the Global Financial Crisis, 1.7x during the Sovereign Debt Crisis, and 2.8x during the COVID-19 pandemic compared to normal periods.

To systematically track this instability, we measure revision volatility (σ_t^{rev}) as the standard deviation of the successive updates made to the initial "flash" GDP estimate of quarter t over the subsequent two years. Figure 1 illustrates the temporal evolution of this metric in Spain, confirming that during major crises, statistical agencies are not merely dealing with missing data, but with a structural breakdown of the informational environment.

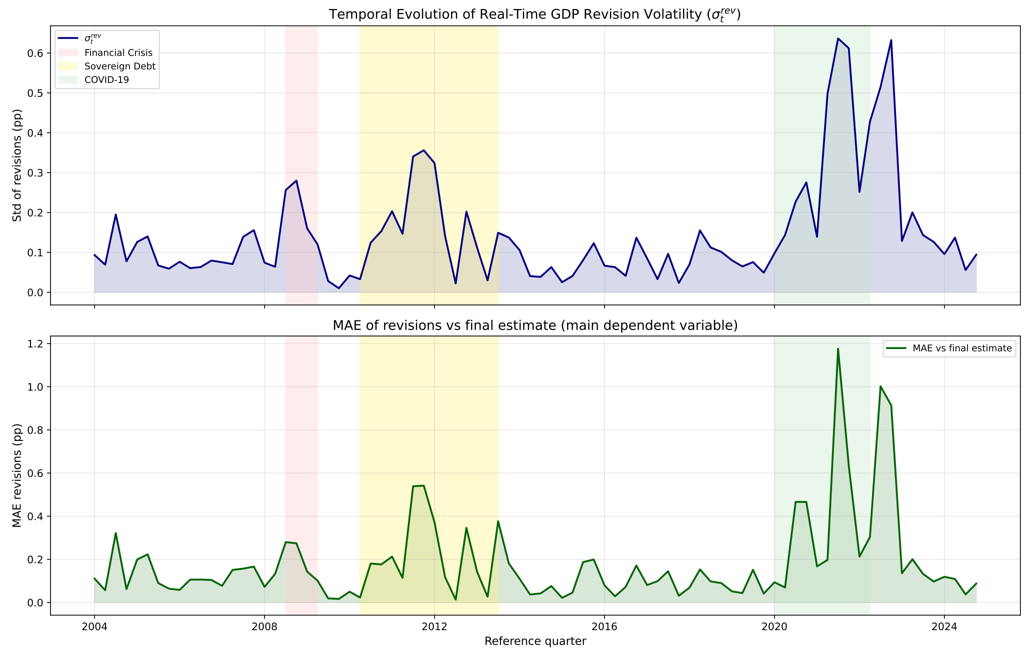

Figure 1: Temporal Evolution of Real-Time GDP Revision Volatility (σ_t^{rev}) in Spain (2005–2024). The shaded areas represent major crisis episodes: the Global Financial Crisis (2008–2009), the European Sovereign Debt Crisis (2010–2012), and the COVID-19 pandemic (2020–2022).

Despite the well-documented existence of these revision spikes (Pavía et al., 2018; Croushore, 2011), the precise mechanism through which economic uncertainty translates into statistical error remains largely unmeasured. The existing literature is divided into two distinct traditions: the *revision literature*, which asks how much and when data is updated (Aruoba, 2008; Asimakopoulos et al., 2023), and the *uncertainty measurement literature*, which seeks to quantify economic ambiguity using various proxies like the VIX (Bloom, 2009), text-based indices (Baker et al., 2016), or unpredictable volatility (Jurado et al., 2015). This paper bridges these two fields by formalizing the link between the entropic state of a forecasting system and the quality of official statistics.

We hypothesize that the magnitude of revision errors reflects the informational disorder prevalent at the time of the initial estimate. Rather than pursuing a formal information-theoretic proof, our approach is strictly operational: we measure this disorder by aggregating three observable dimensions of predictive ambiguity within a forecasting ensemble. These dimensions capture average internal model dispersion, disagreement among competing modelling paradigms, and the instability of forecasts between consecutive periods. The theoretical motivation for this decomposition is outlined in Appendix A.

Our contribution is primarily empirical. First, we provide a framework that maps the internal statistics of a forecasting ensemble to observable, real-time measures of predictive ambiguity, functioning entirely without look-ahead bias. Second, we extend the analysis of National Accounts quality in Spain (Pavía et al., 2018), documenting the evolution of revision patterns through recent crisis episodes. Third, we establish the CEUI as a leading indicator of data-quality variation that improves on standard macroeconomic uncertainty proxies. Over the 2015Q4–2024Q4 evaluation window ($N = 37$ quarters), the Spearman rank correlation between the index and revision volatility is $\rho = 0.729$ ($p < 0.001$). Estimating our baseline specification yields a positive slope, $\hat{\beta} = 0.0199$, significant at the 1% level, with $R^2 = 0.417$. In a horse-race against the VIX and the Economic Policy Uncertainty (EPU) index for Spain, the CEUI carries predictive information about data quality that the external proxies do not: it remains significant at the 1% level while the VIX and EPU become insignificant once it is included.

The results suggest that the dispersion across models and the instability of revisions observed in institutional forecasting are not merely noise; they are informative signals about the underlying reliability of real-time macroeconomic data. For producers and users of statistics, our index provides an ex-ante measure of the level of caution that should be applied to current estimates before committing to major economic decisions.

The remainder of the paper is structured as follows. Section 2 reviews the literature on data revisions and uncertainty measurement. Section 3 describes the real-time data vintages, the target variable, and the revision process. Section 4 outlines the construction of the real-time uncertainty index. Section 5 details our empirical strategy. Section 6 presents the main results—the association with GDP revisions and the horse-race against external benchmarks. Section 7 provides robustness checks, Section 8 discusses the findings, and Section 9 concludes.

2 Literature Review and Methodological Context

The literature on data revisions has traditionally focused on characterizing the statistical properties of noise and news in economic series. Mankiw and Shapiro (1986) famously formalized the distinction between "news" (where revisions reflect new information and are unpredictable, meaning preliminary estimates are rational forecasts) and "noise" (where revisions represent measurement errors and are predictable). This dichotomy has been tested extensively: Mankiw et al. (1984) and Kavajecz and Collins (1995) analyzed preliminary releases of monetary aggregates and detected notable departures from rationality, while Mork (1987) challenged early GNP forecasts, showing they contained systematic errors. Historically, the importance of this statistical friction was recognized by Howrey (1978), who documented that using preliminary data in econometric models leads to biased forecasts. In international contexts, Faust et al. (2005) evaluated G-7 countries and showed that early GDP releases are highly predictable, while Fixler and Grimm

(2002) highlighted the recurring reliability challenges of US GDP estimates, especially at cyclical turning points. Within Spain, Pavía et al. (2018) provided a comprehensive evaluation of GDP quality, revealing systematic seasonal and regime-dependent patterns of revision. More recently, Asimakopoulos et al. (2023) has emphasized that the trade-off between timeliness and accuracy is increasingly critical, as statistical agencies rush early estimates under severe constraints. These errors are fundamentally linked to the statistical production process; as Chen et al. (2018) shows, reconciling different sectoral accounts (income, expenditure, and production) is a complex, lagged process that mechanically triggers revisions. However, as Aruoba (2008) demonstrates, revision errors are not "well-behaved"—they exhibit persistent, heteroscedastic characteristics that standard state-space models fail to absorb. This suggests that the quality of official statistics is endogenous to the level of informational disorder in the macroeconomic environment.

Parallel to the revision literature, the quantification of economic uncertainty has seen rapid advancement, moving from simple market-based measures to complex multi-variable systems. Bloom (2009) pioneered using stock market volatility (VIX) to capture uncertainty shocks. To capture a broader macroeconomic footprint, Jurado et al. (2015) defined uncertainty as the common unpredictable component of a large panel of economic series, establishing a conditional volatility benchmark. Beyond financial and statistical volatility, Baker et al. (2016) introduced the Economic Policy Uncertainty (EPU) index, leveraging newspaper text, tax expirations, and forecaster disagreement. In the Spanish context, Ghirelli et al. (2019) constructed a localized Spanish EPU, linking uncertainty directly to regional and national policy shocks. Despite these achievements, these measures capture generalized risk and often overlook statistical uncertainty. In this regard, Berge (2021) demonstrated that real-time output gap estimates are subject to time-varying uncertainty that fluctuates with the business cycle. Furthermore, standard indicators ignore the epistemic uncertainty originating from forecasting systems. Measuring this uncertainty through forecaster disagreement has a long history: Mankiw et al. (2004) documented massive disagreement in inflation expectations among professional and household forecasters, showing it co-varies with the business cycle. Extending this concept, Lahiri and Sheng (2010) provided the theoretical link showing that overall forecast uncertainty is composed of the variance of future shocks and the cross-sectional disagreement among forecasters, establishing disagreement as a valid proxy for epistemic ambiguity. Nonetheless, relying on subjective surveys, as in Clark and Mertens (2017) or Carriero et al. (2018), introduces behavioral and structural limitations, leaving a gap for objective, model-based indicators.

The central gap in the existing literature—the "missing link"—is the formal connection between ex-ante uncertainty and ex-post statistical quality. While it is intuitive that revisions should be larger during crises, the literature lacks a unified metric that is systematically associated with this breakdown. Conceptually, this problem stems from the classical distinction between risk (measurable randomness) and uncertainty (unmeasurable randomness) outlined by Knight (1921). When the economic environment degrades, forecasting models transition from simple risk to what Ellsberg (1961) defined as ambiguity, where the underlying probability distributions are unknown and models diverge. Our framework bridges this gap by introducing informational entropy as a sufficient statistic for the state of the forecasting system. This approach has key historical precursors: Patterson and Heravi (1991) first applied Shannon's informational entropy to characterize the reduction in uncertainty as national accounts are revised over time, while Shoja and Soofi (2017) demonstrated that entropy-based measures are superior to standard

Table 1: Revision Error Statistics by Quarter (2004–2024)

Quarter	Mean Error	MAE	Obs
Q1	-0.012	0.123	21
Q2	0.005	0.111	21
Q3	0.045	0.266	21
Q4	-0.007	0.226	21
Average	0.008	0.182	84

variances for capturing the disagreement of economic forecasters. Methodologically, our construction of a multi-model ensemble to measure this disorder is closely related to recent advances in forecast combinations under uncertainty, such as the entropy-based machine learning algorithm for forecast combination proposed by Bretó et al. (2019). By decomposing total entropy into within-model variability, between-model dispersion, and temporal instability, we provide a real-time, objective proxy for the shifting variance of the official data. Unlike unidimensional measures, our CEUI exhibits a strong contemporaneous and lagged association with the magnitude of data revisions, offering a robust and comprehensive uncertainty capture.

3 Data and the Revision Process

3.1 CNTR Vintage Dataset

Our primary dataset for the revision analysis is the real-time database of Spain's *Contabilidad Nacional Trimestral* (CNTR), obtained from the INE. The dataset comprises a vintage triangle of quarterly observations, with publication vintages identified by their exact release timestamps. Each vintage corresponds to an official CNTR publication and contains the full historical GDP level series at that point in time.

The target variable is the quarter-on-quarter GDP growth rate (seasonally adjusted), computed as the percentage change in the GDP level index relative to the preceding quarter. All revision metrics are computed on this growth rate series. We define two operationally precise concepts. The **flash estimate** ($y_{t,\text{flash}}$) is the first vintage published after the reference quarter t has ended. The **final estimate** ($y_{t,\text{final}}$) is the most recent vintage available for each quarter. The **revision** for quarter t is defined as $r_t = y_{t,\text{final}} - y_{t,\text{flash}}$. For the cross-sectional volatility analysis, we define the **revision standard deviation** σ_t^{rev} as the standard deviation of all available vintage estimates for quarter t. This metric uses the full information content of the vintage triangle and is the key dependent variable in our regressions.

3.2 Replication and Extension of Pavía et al. (2018)

Table 1 presents revision error statistics for the 2004–2024 period, computed over a two-year window from each quarter's first publication using the harmonised `cntr2` vintage triangle (base year 2015, 106 vintages). The seasonal pattern identified by Pavía et al. (2018) is broadly confirmed, though Q3 now exhibits the highest MAE (0.266 pp), reflecting the concentration of large upward revisions during the COVID-19 recovery phase when the INE progressively incorporated new information about the rebound in activity.

Table 2: Crisis Amplification of Revision Errors

Period	MAE	Obs	Amplif. Factor
Normal	0.134	56	1.0x
Financial Crisis	0.200	4	1.5x
Sovereign Debt	0.227	14	1.7x
COVID-19	0.379	10	2.8x

Table 2 documents revision amplification across three crisis episodes relative to normal periods. The COVID-19 pandemic produces the largest amplification factor (2.8x), reflecting the unprecedented volatility of that period, while the sovereign debt crisis and the financial crisis produce factors of 1.7x and 1.5x respectively.

The amplification factors documented above establish the empirical pattern that motivates our framework. The remainder of the paper investigates the mechanism: whether the internal disorder of the forecasting system co-varies systematically with the magnitude of these revisions. While the historical revision evidence in Table 2 spans two decades (2004–2024), the CEUI is constructed from 2015Q4 onwards, reflecting the start of our rolling evaluation window. Consequently, the direct validation of the link focuses on the period 2015Q4–2024Q4, which nonetheless encompasses the COVID-19 episode and the post-crisis structural adjustment.

3.3 Forecast Ensemble Descriptive Statistics

To construct our uncertainty index, we employ a five-model ensemble forecasting Spanish GDP: a Vector Autoregression (VAR), a Random Forest (RF), an automatic ARIMA model, a Long Short-Term Memory (LSTM) network, and a Dynamic Factor Model (DFM). These models are estimated on a comprehensive set of 11 predictor variables covering labor market dynamics, industrial activity, and services.

Table 3 presents the descriptive statistics and the Pearson correlation matrix of the forecasts generated by the ensemble over the evaluation period. Panel A documents the mean, standard deviation, minimum, and maximum of the realized GDP and individual model forecasts, while Panel B reveals the correlation structure between the forecasts. The pairwise correlations between forecasts range from -0.009 to 0.871. While not universally high, the positive associations between the most accurate models indicate a shared signal during major cyclical movements, which dominates model-specific idiosyncratic variations. Crucially, the models exhibit significant behavioral heterogeneity under stress: while some adapt quickly and track the sharp movements in GDP, others diverge or lag. This heterogeneous response to incoming data is precisely what motivates the use of between-model dispersion as a measure of epistemic uncertainty.

4 The Real-Time Uncertainty Index

We interpret the forecasting ensemble as an information-processing system. When the economic environment is stable, models converge and errors remain bounded; when a structural shock occurs, the system's internal disorder increases. The information-theoretic motivation is given in Appendix A; here we describe the three observable, real-time measures that the framework maps onto. At each quarter t we compute:

Table 3: Ensemble Descriptive Statistics and Model Correlations

Panel A: Summary Statistics (QoQ % Growth)					
Variable	Mean	Std. Dev.	Min	Max	Obs.
GDP Realized	2.26	5.80	-21.45	19.77	40
VAR Forecast	1.14	15.86	-64.62	40.91	40
DFM Forecast	1.68	5.19	-22.79	14.29	40
ARIMA Forecast	1.38	9.20	-44.71	16.68	40
Random Forest	2.02	3.46	-8.03	8.94	40
LSTM Forecast	2.57	3.47	-7.90	8.21	40
CEUI Index (Raw)	4.07	5.74	0.25	18.88	37

Panel B: Pearson Correlation Matrix between Model Forecasts					
	VAR	RF	ARIMA	LSTM	DFM
VAR	1.000	–	–	–	–
RF	0.127	1.000	–	–	–
ARIMA	0.664	0.578	1.000	–	–
LSTM	0.096	0.787	0.609	1.000	–
DFM	0.498	0.819	0.812	0.791	1.000

Note: Panel A reports quarterly growth statistics. Panel B shows the Pearson correlation matrix between the forecasts of the five models in the ensemble.

1. **Within-model variability** ($\mathcal{U}_t^{\text{within}}$): measures the average uncertainty of individual models about their own forecasts. Under Gaussianity, this maps to the average log-predictive standard deviation, which is directly observable from the confidence intervals produced by each model.

2. **Between-model dispersion** ($\mathcal{U}_t^{\text{between}}$): measures the disagreement among competing modeling paradigms. It is proportional to the empirical standard deviation of the point forecasts across the five models.

3. **Temporal instability** ($\mathcal{U}_t^{\text{temporal}}$): captures the rate at which beliefs are updated between consecutive periods. It is computed as the trailing volatility of the ensemble's forecast path.

Each measure uses only information available up to quarter t, ensuring strict real-time validity.

4.1 The Composite Economic Uncertainty Index (CEUI)

We construct the CEUI directly from the raw standard deviations of the three dimensions. At each quarter t, we observe $\mathcal{U}_t^{\text{within}}$, $\mathcal{U}_t^{\text{between}}$, and $\mathcal{U}_t^{\text{temporal}}$. Because we seek an index that is strictly valid in real time, avoiding any look-ahead bias that arises from full-sample normalisation, we define the baseline CEUI as the simple average of the raw dimensions:

$$\text{CEUI}_t = \tfrac{1}{3}\mathcal{U}_t^{\text{within}} + \tfrac{1}{3}\mathcal{U}_t^{\text{between}} + \tfrac{1}{3}\mathcal{U}_t^{\text{temporal}}. \tag{1}$$

By operating on the raw scale, the index reflects the absolute magnitude of percentage-point deviations in the forecast system, and—because every dimension is backward-looking—it is available in real time. This raw measure is used for all empirical evaluations. For illustration only, when graphing the historical evolution of the index we also report a normalised 0–100 version based on full-sample percentiles; this is strictly an ex-post descriptive transformation (see Appendix B). Figure 2 displays the path of the three individual dimensions of uncertainty and the composite index over the sample period.

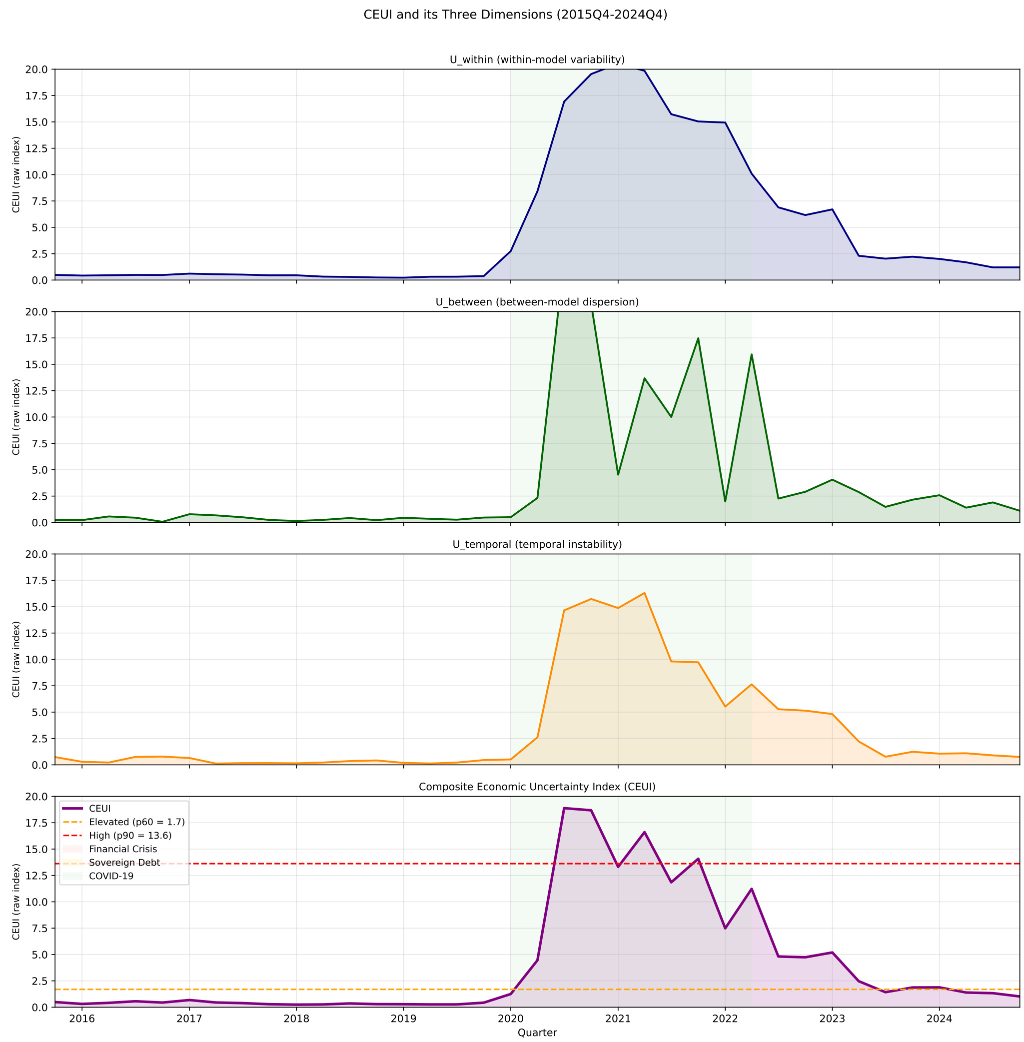

Figure 2: Historical Path of the Three Uncertainty Dimensions and the Raw Composite Economic Uncertainty Index (CEUI) over 2015Q4–2024Q4.

5 Empirical Strategy

Our objective is to test whether the internal disorder of the forecasting ensemble (CEUI_t) predicts the magnitude of the subsequent revision in the official GDP figure (σ_t^{rev}), which is the standard deviation of the revisions accruing to the quarter-t estimate over the following two years. This two-year window is chosen to capture the bulk of the National Accounts regular revision cycle, which typically finalizes quarterly estimates after eight quarters.

5.1 Baseline Specification

We estimate the baseline relationship by Ordinary Least Squares (OLS) with heteroscedasticity-robust (HC3) standard errors:

$$\sigma_t^{rev} = \alpha + \beta \cdot \text{CEUI}_t + \varepsilon_t. \tag{2}$$

Given our relatively small sample size ($N = 37$) and the presence of heavy-tailed observations during crisis periods, we rely on HC3 standard errors for inference in OLS, supplemented by the non-parametric Spearman rank correlation to ensure results are not driven solely by extreme outliers. Because the CEUI is calculated using data available strictly up to quarter t, it is a valid leading indicator for the revisions that unfold over the subsequent two years.

5.2 Horse-Race Specification

To determine whether the CEUI captures information beyond standard uncertainty metrics, we run a horse-race against the VIX (financial-market volatility) and the Economic Policy Uncertainty (EPU) index for Spain:

$$\sigma_t^{rev} = \alpha + \beta_1 \text{CEUI}_t + \beta_2 \text{VIX}_t + \beta_3 \text{EPU}_t + \varepsilon_t. \tag{3}$$

All predictors are standardised to mean zero and unit variance to allow direct comparison of the coefficients.

6 Results

6.1 Association between Index and Revisions

We find a strong, positive association between the real-time CEUI and the magnitude of subsequent revisions. Over the 2015Q4–2024Q4 evaluation window ($N = 37$ quarters), the Spearman rank correlation between the index and revision volatility is $\rho = 0.729$ ($p < 0.001$). Estimating the baseline specification in Equation (2) yields a positive slope, $\hat{\beta} = 0.0199$ (SE ≈ 0.0064), which is significant at the 1% level under HC3-robust standard errors, explaining a substantial portion of the variance ($R^2 = 0.417$). Because the CEUI is calculated using data available strictly up to quarter t, this finding establishes the index as a valid leading indicator for revision volatility that will unfold over the subsequent two years.

To put the magnitude of this effect into context, consider the full observed range of the index, which spans from a minimum of approximately 0.25 during calm periods to a peak of 18.88 at the height of the COVID-19 pandemic. Over this range, the estimated slope implies a total shift of roughly 0.37 percentage points ($\approx 0.0199 \times 18.63$) in revision volatility. This is an economically meaningful effect when compared to the baseline level of σ^{rev} during quiet quarters (around 0.1 pp) versus the highly turbulent COVID-19 quarters (around 0.6 pp).

Figure 3 displays this relationship graphically. While the scatter plot shows a largely linear upward trend, one might be concerned that the result is entirely driven by the cluster of high-uncertainty observations during the pandemic. However, the sensitivity analysis reported in Table 4 confirms that the positive slope is robust to sample selection. The association survives the complete exclusion of the core COVID-19 shock

($\hat{\beta} = 0.0297, p < 0.001$) and remains highly significant when omitting the single most influential observation, 2020Q3 ($\hat{\beta} = 0.0239, p < 0.01$). Thus, the informational entropy of the forecasting system provides a consistent signal of data unreliability across both normal and crisis periods.

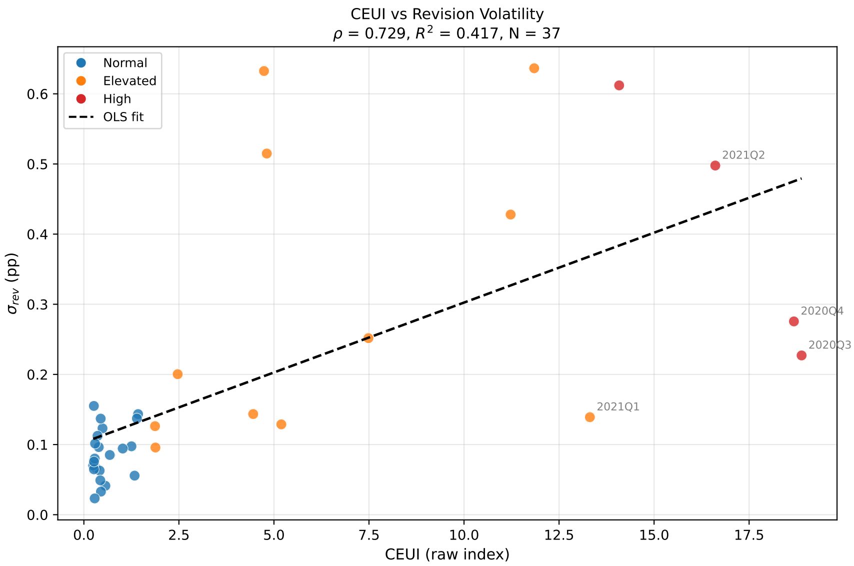

Figure 3: Real-time CEUI versus the volatility of subsequent GDP revisions, 2015Q4–2024Q4 ($N = 37$). The fitted line corresponds to the baseline OLS specification in Equation (2).

Table 4: Sensitivity of Baseline Regression across Subsamples

Subsample	β	t	p	R^2	ρ	N
Full sample	0.0199	3.130***	0.0017	0.417	0.729	37
Ex-COVID (2020Q2-Q4)	0.0297	4.153***	0.0000	0.575	0.686	34
Ex-crisis (2020Q1-2021Q1)	0.0354	5.402***	0.0000	0.701	0.679	32
Ex-most influential (2020Q3)	0.0239	3.239***	0.0012	0.488	0.717	36

Note: HC3-robust standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.2 Horse-Race against VIX and EPU

Table 5 reports the horse-race regressions. We estimate Equation (3) over the overlapping sample for which the CEUI, the VIX and the Spanish EPU index are jointly available ($N = 26$, 2015Q4–2022Q1), standardising all predictors. Estimated individually, the CEUI is the strongest of the three predictors of revision volatility. While the VIX and EPU are powerful indicators of generalized risk (capturing market and political turbulence respectively), they often fail to predict revision volatility because they do not reflect the internal disorder of the statistical production system. In the individual models, the VIX and EPU explain only a small fraction of the variance in revision volatility ($R^2 = 0.057$ and $R^2 = 0.041$, respectively). By contrast, including the CEUI dramatically improves the model's explanatory power, driving the R^2 up to 0.570 when all three are combined.

In this joint specification, the CEUI remains highly significant, whereas the VIX and the EPU index become insignificant. The internal disorder of the forecasting ensemble therefore carries information about the reliability of official data that is not subsumed by market-based or news-based uncertainty measures—the paper's central empirical claim.

Table 5: Horse-Race Regressions: CEUI vs VIX vs EPU as Predictors of Revision Volatility

	CEUI		VIX		EPU		R^2	N
Specification	$\hat{\beta}$	t	$\hat{\beta}$	t	$\hat{\beta}$	t		
(1) CEUI only	0.1212***	2.97	–	–	–	–	0.543	26
(2) VIX only	–	–	0.0392	1.49	–	–	0.057	26
(3) EPU only	–	–	–	–	0.0333	1.11	0.041	26
(4) CEUI + VIX	0.1314***	3.07	-0.0219	-1.18	–	–	0.557	26
(5) CEUI + EPU	0.1354***	3.18	–	–	-0.0303	-1.05	0.569	26
(6) All three	0.1352***	3.08	0.0026	0.11	-0.0323	-1.01	0.57	26

Note: Dependent variable σ_t^{rev}. All predictors standardised (zero mean, unit variance). HC3-robust t-statistics. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$. Sample restricted to periods where the CEUI, VIX and EPU are jointly available ($N = 26$, 2015Q4–2022Q1).

7 Robustness Checks

7.1 Real-Time Validity

A common critique of composite indices is look-ahead bias induced by full-sample normalisation, whereby an index value at quarter t depends on observations realised after t. Our baseline CEUI avoids this by construction: it is built from the raw, backward-looking dimensions, each of which uses only information available up to quarter t, and it requires no full-sample rescaling. The headline association ($\rho = 0.729$) is therefore established entirely in real time. The normalised 0–100 transformation used in the descriptive figures (Appendix B) is an ex-post device that enters none of the estimates reported above; and because the central result is a rank correlation, it is in any case invariant to any monotonic rescaling of the index. As a more demanding check, replacing the raw dimensions with their full-sample percentile-normalised counterparts—a transformation that does alter the composite's ranks—still yields $\rho = 0.729$, identical to the raw-index value and confirming that the central finding is not an artefact of the scaling choice.

7.2 Alternative Weighting Schemes

Table 6 presents CEUI variants constructed under four alternative weighting schemes for the three uncertainty dimensions.

All alternative schemes achieve correlations above 0.997 with the equal-weight baseline and regime concordance above 97%. These results confirm that the CEUI is virtually invariant to the choice of weights, primarily because the three dimensions share the dominant crisis signal. Equal weighting is retained as the baseline on grounds of parsimony and transparency.

Table 6: Robustness to Alternative Weighting Schemes

Scheme	w_{within}	w_{between}	w_{temporal}	Corr. baseline	Peak	Concordance
Equal (Baseline)	0.333	0.333	0.333	1.000	96.30	100.0%
Dispersion-heavy	0.200	0.500	0.300	0.997	96.67	97.3%
Volatility-heavy	0.500	0.300	0.200	0.997	95.83	100.0%
Temporal-heavy	0.200	0.300	0.500	0.997	96.67	100.0%

Note: Corr. baseline is the Pearson correlation with the equal-weight CEUI. Concordance is the percentage of periods where the regime classification (Normal, Elevated, High) matches the baseline. All correlations > 0.99.

7.3 Leave-One-Model-Out Analysis

To ensure that the uncertainty signals are not driven by a single outlier model, we conduct a leave-one-model-out (LOO) analysis. By systematically excluding one model from the ensemble and recomputing the CEUI, we verify the stability of the aggregate index. As detailed in Table 7, the LOO indices maintain very high correlations with the full ensemble, ranging from 0.818 to 0.841. Moreover, the predictive link with revision volatility is remarkably stable: the maximum change in the Spearman correlation ($\Delta\rho$) is merely -0.029 when the VAR model is excluded.

Table 7: Leave-One-Model-Out (LOO) Ensemble Stability

Excluded model	Corr. with full ensemble	Δ Peak	$\Delta\rho_{\text{Spearman}}$
VAR	0.818	+77.415	-0.029
ARIMA	0.839	+77.415	+0.017
RF	0.841	+78.341	+0.002
LSTM	0.841	+78.341	-0.020
DFM	0.840	+77.415	+0.004

Note: $\Delta\rho_{\text{Spearman}}$ measures the stability of the predictive link with revision volatility when one component is removed. The results confirm that no single model disproportionately drives the aggregate uncertainty signal.

The LOO analysis demonstrates that the structural features of the CEUI—particularly the crisis peaks and the correlation with official revisions—remain stable regardless of the specific composition of the model ensemble.

7.4 Normalization Sensitivity

Our baseline implementation uses raw dimensions for regression and percentile ranks to normalize the uncertainty components for graphical illustration. Table 8 compares this approach with alternative scaling methods, documenting the consistency of the resulting index. While the continuous correlations with the min-max baseline are high (0.841 for percentile rank and 1.000 for Z-score), the discrete regime agreement varies significantly (from 18.9% to 59.5%), reinforcing our choice to rely on the raw, unscaled index for the core empirical regressions rather than on discrete regimes.

Table 8: Sensitivity to Normalization Methods

Method	Corr. vs min-max	Regime agreement (%)
Percentile rank (baseline)	0.841	18.9%
Z-score	1.000	59.5%
Min-max [0,100]	1.000	100.0%

Note: Regime Agreement measures the consistency of high-uncertainty flags across methods relative to the Min-Max baseline.

The high degree of regime agreement across normalization techniques suggests that the detection of extreme uncertainty episodes is more a function of the underlying data dynamics than the transformation applied.

8 Discussion

The empirical evidence supports a simple reading: when a panel of heterogeneous models disagrees and becomes unstable in real time, the official GDP figure released in that quarter is more likely to be substantially revised later. The mechanism driving this predictive power likely lies in a shared deterioration of the underlying informational environment. When primary, high-frequency data streams enter in a disorderly or conflicting manner, the forecasting models, which weigh inputs differently, naturally diverge. Simultaneously, the statistical agency's own triangulation of early data becomes inherently more fragile, leading to flash estimates that are prone to larger subsequent corrections once comprehensive survey data become available.

This finding bridges two distinct strands of literature. On one hand, Aruoba (2008) documents that GDP revisions are often heteroscedastic and not "well-behaved," implying that data uncertainty varies over time. On the other hand, macroeconomic uncertainty measurement—exemplified by Jurado et al. (2015) and Baker et al. (2016)—focuses on capturing periods when the economy becomes fundamentally harder to predict. The CEUI acts as a missing link between ex-ante forecasting uncertainty and ex-post statistical quality, showing that the internal entropy of the forecasting system provides a direct real-time proxy for the shifting variance of the data itself.

From a practical standpoint, the index offers a readily deployable tool for producers and users of statistics. Because the signal is available immediately and is built entirely from models an agency might already run internally, it does not depend on costly external financial data or complex text-mining algorithms. A statistical agency could use a high reading of the CEUI as an internal "cautionary traffic light," signalling that the flash GDP estimate is unusually fragile and that policymakers should perhaps delay irreversible decisions until further data consolidates.

8.1 Model Resilience

A secondary observation concerns model behaviour under extreme shocks. Linear models such as ARIMA, accurate in normal times, see their errors deteriorate sharply during crises, whereas the non-linear machine-learning components retain more adaptive capacity. This heterogeneity is what makes between-model dispersion informative; details are reported in Appendix C.

8.2 Limitations

Two qualifications shape the interpretation. First, the index predicts the *volatility* of revisions, not their *sign*: it flags fragile vintages, not the direction of the future correction. Furthermore, our evaluation window spans 2015Q4–2024Q4, yielding a small sample ($N = 37$). We mitigate this with heteroscedasticity-robust standard errors, but caution is warranted in extrapolation. The dominance of the COVID-19 pandemic in the sample means the index is heavily tested against one unprecedented shock. Additionally, the relationship could reflect a common confounding factor, where extreme macroeconomic volatility directly causes both forecast disagreement and measurement difficulties, rather than a direct causal chain. Finally, there is a recognized gap between theory and measurement: while information-theoretic entropy motivates the index conceptually, the operational measures are built from practical, backward-looking standard deviations.

9 Conclusion

We construct a Composite Economic Uncertainty Index (CEUI) from the internal statistics of a five-model forecasting ensemble for Spanish GDP, computed strictly in real time. The index is strongly associated with the subsequent volatility of official GDP revisions, and a horse-race shows that it carries predictive information beyond the VIX and the Spanish EPU index, both of which become insignificant once the CEUI is included. The measure offers statistical agencies and forecasters a transparent, real-time gauge of the reliability of early GDP estimates, built only from the internal properties of their own forecasting models. Extending the approach to other countries and to the sign of revisions are natural directions for future work.

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A Information-Theoretic Foundations

This appendix gives the information-theoretic motivation for the three operational measures used in the body; the measures themselves are the backward-looking standard deviations defined in Section 3, and the development here is an approximation rather than an exact identity.

Treat the ensemble's quarter-t forecast as a mixture of the five models' predictive densities, $p_t(y) = \frac{1}{M}\sum_{m=1}^{M} p_{m,t}(y)$ with $M = 5$. The Shannon entropy of this predictive distribution,

$$H(p_t) = -\int p_t(y)\log p_t(y)\,dy, \tag{4}$$

is a natural scalar measure of the system's predictive disorder. A standard decomposition writes the entropy of the mixture as the average entropy of the components plus their Jensen–Shannon divergence (Haussler and Opper, 1997; Heskes, 1998):

$$H(p_t) \;=\; \underbrace{\tfrac{1}{M}\sum_{m} H(p_{m,t})}_{\mathcal{U}_t^{\text{within}}} \;+\; \underbrace{\text{JSD}\big(p_{1,t},\ldots,p_{M,t}\big)}_{\mathcal{U}_t^{\text{between}}}. \tag{5}$$

The first term motivates the *within-model* measure (how uncertain each model is about its own forecast) and the second the *between-model* measure (how much the models disagree). To capture the dynamic dimension of uncertainty, we introduce a third component based on the informational distance between consecutive ensemble distributions: the Kullback–Leibler divergence of the current ensemble from the previous period's ensemble:

$$\mathcal{U}_t^{\text{temporal}} = D_{\text{KL}}(p_t \,\|\, p_{t-1}) = \int p_t(y)\log\frac{p_t(y)}{p_{t-1}(y)}\,dy. \tag{6}$$

Under a Gaussian approximation, in which each model m generates a predictive distribution $p_{m,t} = \mathcal{N}(\hat{y}_{m,t}, \sigma^2_{m,t})$, these terms simplify as follows:

Proposition 1 (Operational Equivalences under Gaussianity)**.** *Under Assumption G, the three components of predictive uncertainty satisfy:*

(i) ***Within-model uncertainty:***

$$\mathcal{U}_t^{\text{within}} = \frac{1}{M}\sum_{m=1}^{M} H(\mathcal{N}(\hat{y}_{m,t}, \sigma^2_{m,t})) = \frac{1}{2}\ln(2\pi e) + \frac{1}{M}\sum_{m=1}^{M}\ln\sigma_{m,t} \tag{7}$$

which is a monotone increasing function of the average log-predictive standard deviation.

(ii) ***Between-model dispersion:*** *Under homogeneous variance $\sigma^2_{m,t} = \bar{\sigma}^2_t$ for all m, the Jensen–Shannon divergence satisfies:*

$$\text{JSD}\big(p_{1,t},\ldots,p_{M,t}\big) \approx \frac{1}{2\bar{\sigma}^2_t}\cdot\frac{1}{M}\sum_{m=1}^{M}(\hat{y}_{m,t} - \bar{y}_t)^2 \tag{8}$$

where $\bar{y}_t$ is the cross-sectional mean of point forecasts. The JSD is thus proportional to the cross-sectional variance of point forecasts.

(iii) ***Temporal instability:*** *Considering consecutive Gaussian ensembles* $p_t = \mathcal{N}(\bar{y}_t, \bar{\sigma}_t^2)$ *and* $p_{t-1} = \mathcal{N}(\bar{y}_{t-1}, \bar{\sigma}_{t-1}^2)$*, the KL divergence takes the closed form:*

$$\mathcal{U}_t^{temporal} = \frac{(\bar{y}_t - \bar{y}_{t-1})^2}{2\bar{\sigma}_{t-1}^2} + \frac{1}{2}\left(\frac{\bar{\sigma}_t^2}{\bar{\sigma}_{t-1}^2} - 1 - \ln\frac{\bar{\sigma}_t^2}{\bar{\sigma}_{t-1}^2}\right) \tag{9}$$

This establishes that our operational measures (standard deviations and forecast differences) are direct proxies to theoretical entropy components.

B Uncertainty Regimes

While our main empirical results use the raw CEUI, classifying periods into discrete uncertainty regimes provides illustrative operational context. For this purpose we define the *Normal* regime as CEUI $\leq$ p60, *Elevated* as CEUI $\in$ (p60, p90], and *High* as CEUI $>$ p90 based on the percentiles of the raw index (with full-sample values approximately p60 $\approx$ 1.69 and p90 $\approx$ 13.62). To preserve real-time validity, the percentiles are computed recursively using expanding windows. The 0–100 scale is used only for the illustrative figure, not for the tables.

Table 9 shows the descriptive properties of the regimes, and Table 10 reports the stability of the thresholds under a bootstrap analysis. Figure 4 shows the evolution of the CEUI and the uncertainty regimes.

Table 9: Characteristics of Revisions by Uncertainty Regime

Regime	CEUI range	N	N_{rev}	MAE (pp)	vs Normal
Normal	$<=$ 1.7	22	22	0.087	1.0x
Elevated	1.7-13.6	11	11	0.3	3.5x
High	$>$ 13.6	4	4	0.403	4.7x

Table 10: Bootstrap Threshold Stability Analysis (n = 1,000)

Threshold	Baseline	Boot mean	Boot std	95% CI	
				2.5%	97.5%
Elevated (p60)	1.69	2.11	1.21	0.64	4.78
High (p90)	13.62	13.29	3.11	5.19	18.68

C Model Specifications and Resilience

The ensemble comprises five models estimated under an identical rolling protocol:

1. **Vector Autoregression (VAR):** We implement a VAR model with 4 lags and 11 variables:

$$\mathbf{Y}_t = \mathbf{c} + \sum_{i=1}^{4} \mathbf{A}_i \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t \tag{10}$$

where $\mathbf{Y}_t$ is the vector of endogenous variables and $\boldsymbol{\varepsilon}_t \sim N(0, \boldsymbol{\Sigma})$.

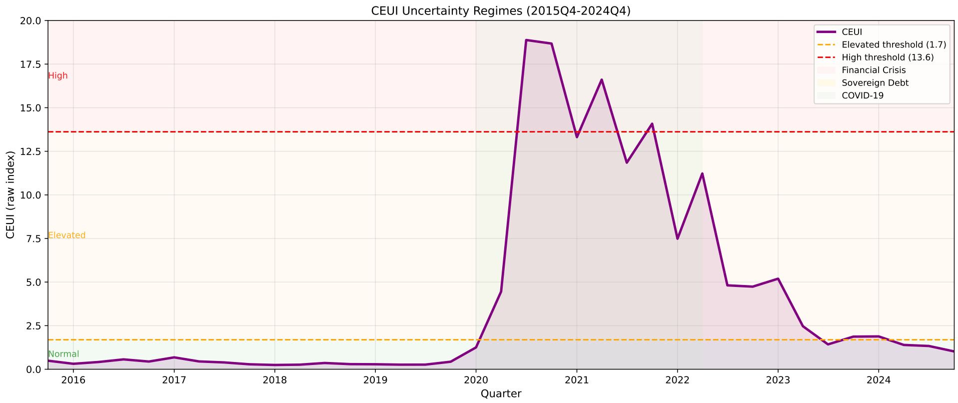

Figure 4: CEUI evolution and recursive uncertainty-regime thresholds, 2015Q4–2024Q4.

2. **Random Forest (RF):** The Random Forest implementation uses 50 trees with maximum depth of 3, minimum samples per split of 2, and rolling window estimation.

3. **ARIMA:** We employ automatic ARIMA specification with model selection based on information criteria, testing combinations of (p, d, q) parameters up to order 3.

4. **Long Short-Term Memory (LSTM):** The LSTM architecture employs 6-quarter input sequences with two LSTM layers (24 and 24 units), dropout regularization (20%), and dense output layer.

5. **Dynamic Factor Model (DFM):** We implement a DFM with 2 common factors estimated using Kalman filtering and maximum likelihood estimation:

$$\mathbf{Y}_t = \mathbf{\Lambda}\mathbf{F}_t + \boldsymbol{\varepsilon}_t, \quad \mathbf{F}_t = \mathbf{A}(L)\mathbf{F}_{t-1} + \boldsymbol{\eta}_t \tag{11}$$

Table 11 compares the models' forecast accuracy in normal versus crisis periods, and Figure 5 shows the ratio of crisis to normal-period forecast error.

Table 11: Model Performance Summary (2019Q1–2024Q4)

Model	MAE	RMSE	MAE pre-COVID	MAE post-COVID	MAE ratio	Resilience
VAR	7.102	15.469	0.519	13.685	26.39x	0.038
ARIMA	4.462	8.615	0.239	8.684	36.30x	0.028
RF	2.176	4.920	0.384	3.968	10.33x	0.097
LSTM	2.146	4.777	0.612	3.680	6.02x	0.166
DFM	2.417	5.382	0.287	4.547	15.83x	0.063

D Influence Diagnostics

We report influence diagnostics for our baseline regression. Table 12 presents the diagnostic statistics, and Figure 6 displays the influence diagnostics. The results confirm that the positive association between the CEUI and revision volatility is not driven by any single observation: the maximum Cook's distance is $D = 0.556$ for 2020Q3.

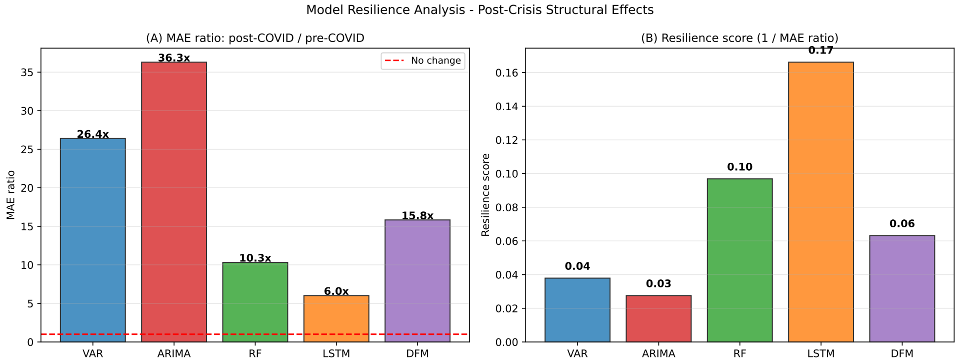

Figure 5: Model resilience: ratio of crisis to normal-period forecast error by model.

Table 12: Diagnostic Statistics for Influence Analysis

Quarter	CEUI	σ_{rev}	Cook's D	Leverage	Std. Resid.	Flagged
2020Q3	18.88	0.2270	0.5807	0.2124	-2.0755	Yes
2020Q4	18.68	0.2756	0.3509	0.2073	-1.6380	Yes
2021Q3	11.85	0.6365	0.2165	0.0782	2.2601	Yes
2021Q4	14.08	0.6120	0.1967	0.1117	1.7684	Yes
2021Q1	13.31	0.1390	0.1714	0.0992	-1.7643	Yes
2022Q4	4.74	0.6325	0.1460	0.0274	3.2194	Yes
2022Q3	4.81	0.5149	0.0773	0.0275	2.3380	No
2021Q2	16.61	0.4978	0.0245	0.1599	0.5070	No
2022Q2	11.23	0.4279	0.0221	0.0703	0.7650	No
2017Q4	0.28	0.0233	0.0083	0.0391	-0.6375	No
Threshold: Cook's $D > 4/N = 0.1081$						

E VAR Specification Robustness

The main results are robust to the choice of VAR lag length and specification used in the ensemble. Table 13 compares the baseline VAR(AIC selection, 10 variables) with a parsimonious VAR(2 lags, 4 core variables) model.

Table 13: Robustness to Model Parsimony: Baseline vs Parsimonious VAR

Specification	Avg. parameters	MAE (pp)	$\rho(\text{CEUI}, \sigma_{rev})$
VAR (AIC selection, 10 variables)	410	7.102	0.729
VAR (2 lags, 4 core variables)	36	4.322	-

F Vintage Maturity Control

The association survives controlling for the maturity of each vintage (the number of subsequent releases available), addressing the concern that more recent quarters mechanically display different revision behaviour. Table 14 reports these regressions.

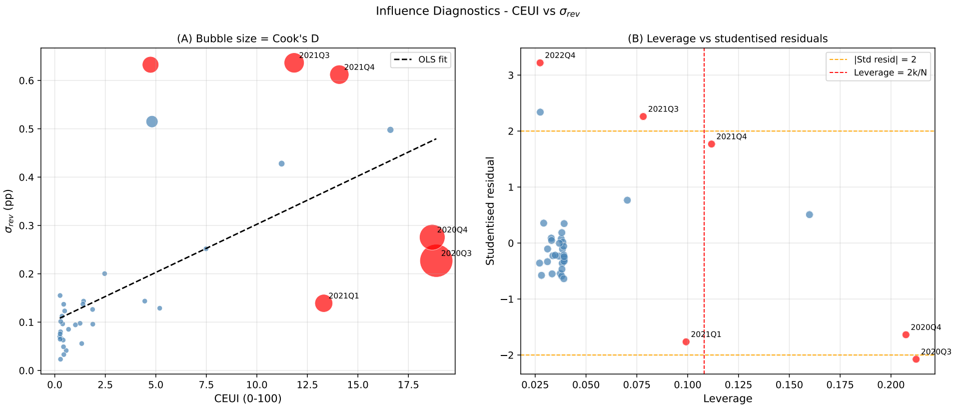

Figure 6: Influence diagnostics for the baseline regression (Equation (2)).

Table 14: Robustness Check: Controlling for Vintage Maturity

Variable	(1) Baseline	(2) Maturity control	(3) Excl. young ($vint < 8$)
CEUI	0.0199***	0.0187***	0.0197***
	(3.13)	(2.89)	(3.09)
vint	–	0.0050	–
		(1.54)	
R^2	0.417	0.427	0.410
N	37	37	35

G Correlation Matrix

The three uncertainty dimensions are highly correlated with one another, which explains the near-invariance of the CEUI to alternative weighting schemes. Figure 7 presents the pairwise Spearman correlations.

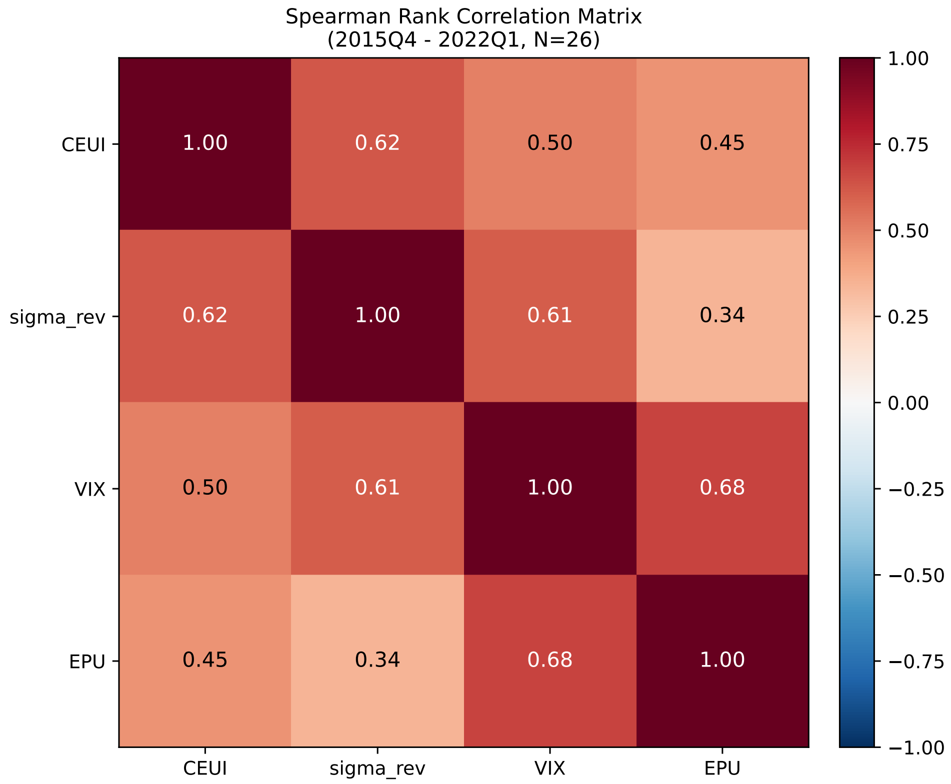

Figure 7: Spearman rank correlations among the CEUI, revision volatility (σ^{rev}), and the external benchmarks (VIX and EPU), over the overlapping horse-race sample ($N = 26$, 2015Q4–2022Q1).